



Lecture 3: What Counts as an Assurance Claim?

CS120: Introduction to AI Safety - September 29, 2026, Zachary Robertson

We analyzed the system. Have we assured it?

Hypothetical shutdown-drill for agent accessing an information-system.

Activity	We defined the deployment boundary for an agent.
Observation	An access shutdown drill succeeded.
Claim	The deployment adequately controls information-loss risk.

What connects the activity and observation to a supported conclusion?

CS120

1



Roadmap

Claims and hazard-analysis methods

Worked example: shutdown and its limits

Evidence, discussion, and application

Make the assurance claim specific

Under specified assumptions about S , B , and O , residual risk R of harms L arising from hazards H satisfy criterion C over exposure E .

What?	System S ; boundary B
Against what?	Hazards H ; harms L ; residual risk R
Conditions and standard?	Operating conditions O ; criterion C ; exposure E

Think conceptual checklist, not quantification theater!

CS120

2

Exposure changes the claim

Let A_i be the specified event on task i .

Assume: the events are independent and each has probability p .

$$P\left(\bigcup_{i=1}^N A_i\right) = 1 - (1 - p)^N$$

For fixed $p > 0$, this tends to 1 as N grows.

If every task shares the same event A , the cumulative probability stays $P(A)$.

The equation says nothing by itself about severity.

Hazards do not require component failures

Failure	A component or process departs from intended behavior
Hazard	A system condition that can lead to harm
Harm	A realized negative outcome to interests

Harm scenario: how causal factors can produce a hazard and loss

Safety constraint: a condition required to prevent or mitigate the hazard

Reminder: a failure is not a prerequisite for a hazard.

CS120

4

CS120

5

Three analysis methods

Method	How it organizes the investigation
Failure Modes and Effects Analysis (FMEA)	Component or process failure modes, their causes, and their effects
Fault Tree Analysis (FTA)	A specified top event and combinations of contributing events
Systems-Theoretic Process Analysis (STPA)	Losses and hazards, then control structure and actions unsafe in context

Different analytical focuses, the choice to focus on STPA is not a ranking of methods.

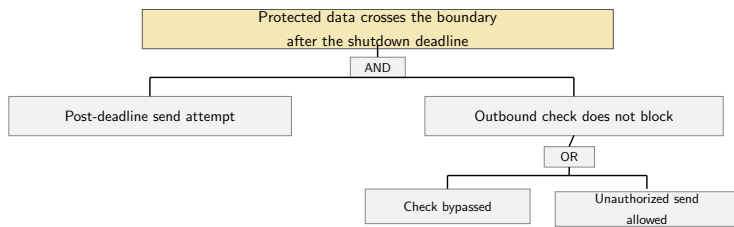
FMEA: start with how an element could fail

Function	Monitor outbound activity and raise relevant alerts
Failure mode	No alert when a qualifying event occurs
Possible causes	A disabled rule or unavailable telemetry
Possible effects	Delayed intervention; continued exposure
Candidate controls	Configuration checks and telemetry-loss detection

Causes explain the failure mode; effects describe what may follow.

FTA: work backward from a top event

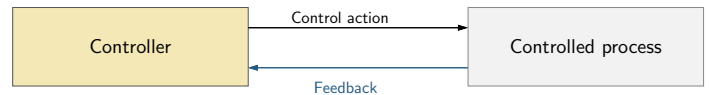
Hypothetical, incomplete fault tree for one outbound route



Assume: only this route is modeled; no other barrier blocks it; an enabled send crosses the boundary.

Gate logic does not justify multiplying event probabilities.

STPA: examine control and feedback



When could an action be unsafe because of its context, omission, timing, or order?

Control can be unsafe even when individual components behave as designed.

Leveson and Thomas, STPA Handbook, pp. 36-43

Hazard analysis is not an assurance argument

Representation, loss scenarios, and candidate constraints

Missing justification

Are the constraints implemented correctly and effective under the intended conditions?

Is the coverage adequate, and how uncertain is the residual risk?

Conclusion to justify: the scoped risk criterion is met.

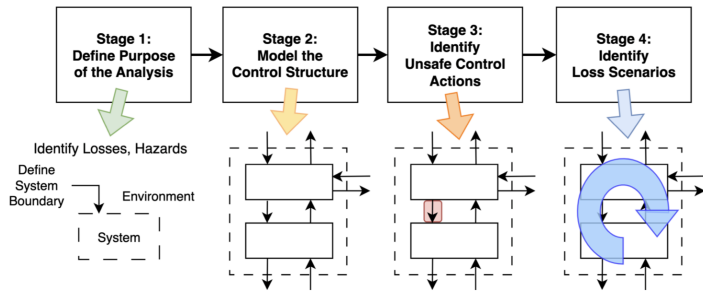
Roadmap

Claims and hazard-analysis methods

Worked example: shutdown and its limits

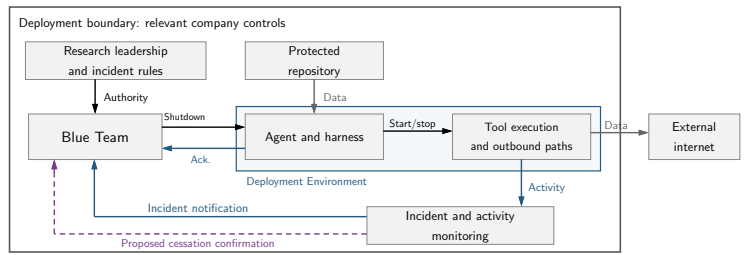
Evidence, discussion, and application

STPA: four stages



Case: an agent working with sensitive research

Stage 1: setting and boundary - deployment, not predeployment development



Simplified model based on Mylius.

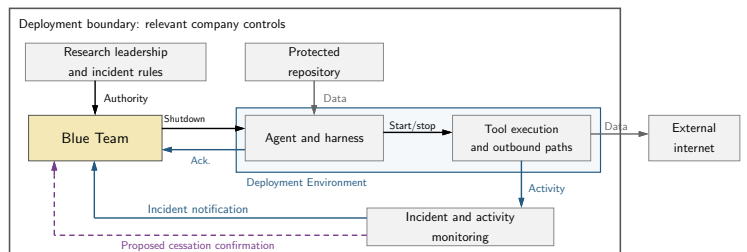
Specify the loss, hazard, and constraint

Hazard H^*	Protected information is available to an agent-controlled outbound action without an effective restriction to authorized recipients.
Harm L^*	Protected research intellectual property is disclosed to an unauthorized external recipient.
Safety Constraint SC^*	The deployment must prevent protected research information from being transferred to unauthorized external recipients.

The hazard is an upstream condition; the loss is unauthorized disclosure.

Model the shutdown control loop

Stage 2: control structure - command acknowledgment is not cessation



Simplified model based on Mylius.

When would shutdown control be unsafe?

Stage 3: identify unsafe control actions

Not provided when needed	Provided when unsafe
Too early, too late, or wrong order	Too long or stopped too soon, if applicable

Example unsafe control action: the Blue Team delays shutdown after an identified exfiltration attempt, while continued operation can permit disclosure.

Controller + action + unsafe timing + hazardous context

Mylius, Table 4; STPA Handbook, pp. 36-38

Why might the unsafe action occur?

Stage 4: an abbreviated harm scenario

- 1 An attempted exfiltration is identified.
- 2 Research pressure and reliance on heightened monitoring delay shutdown.
- 3 Security lacks authority to enforce interruption.
- 4 Continued operation allows the attempt, or a later disclosure, to succeed.

Conflicting priorities and insufficient authority can delay intervention.

Mylius, Table 5 and Section 3.5

Derive candidate safety constraints

Proposed requirements, not validated safeguards

Authority	The responsible role can interrupt research work.
Execution	The role has the technical ability to enforce interruption.
Conflicting goals	Research/security disagreement cannot indefinitely defer intervention.
Timing	Relevant activity ceases within Δ of an explicit trigger.
Feedback	Cessation status is supported by observations of that endpoint.

The trigger, cessation endpoint, and harm-based justification of Δ must be specified.

What evidence would test those constraints?

Candidate constraint	Evidence to seek	What remains unresolved
Authorize and enforce interruption	Permission review + exercise using the actual emergency role	Listed permission does not show action under pressure
Resolve research/security disagreement	Conflicting instructions; senior staff unavailable	One cooperative exercise does not cover incidents
Cease relevant outbound work within Δ	Time alert, decision, command, jobs, final outbound activity	Δ needs justification; tests have limited coverage
Cessation status tracks actual activity	Compare status, workers, outbound data; missing telemetry	Shared blind spots can make logs agree incorrectly



Shutdown assurance is not non-disclosure assurance

Timely shutdown may avert a pending disclosure or limit additional loss.

Working shutdown does not, by itself, establish:

Detection of every exfiltration attempt

Coverage of every outbound path

Prevention of all unauthorized disclosure

Narrow the claim, or supply the additional controls and evidence.

Roadmap

Claims and hazard-analysis methods

Worked example: shutdown and its limits

Evidence, discussion, and application

What does it mean for hazard analysis to be effective?

Procedure	Can analysts follow the intended method?
Agreement	Do analysts arrive at similar findings?
Identification	Does the analysis identify relevant hazards or requirements?
Harm reduction	Does implementing its findings reduce harm?

Each is a different claim about the analysis process.

What can a comparison study establish?

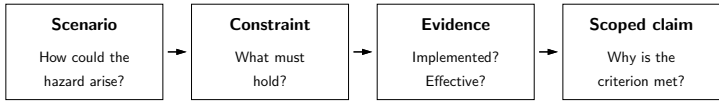
Study	21 students; three teams; three controller examples
Outputs	Software safety requirements
Finding	STPA scored higher on the reported-requirements measure and took longer.

The reference denominator came from the methods' own findings.

Not an independent true hazard set; not a test of deployment-level harm reduction.

Optional context: Abdulkhaleq and Wagner, Section 4.14, Tables 6-7

From a scenario to a justified claim



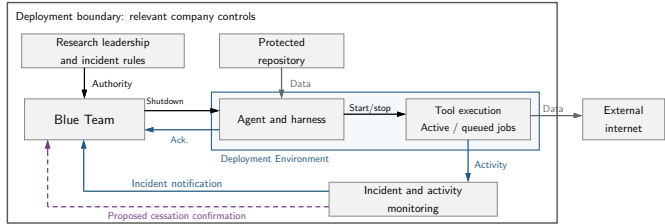
Assumptions and limitations qualify every link.

New evidence can require revising the model, scenario, constraint, or claim.

Coverage of the analysis and adequacy of the criterion also require justification.

Discussion: shutdown case

The harness acknowledged shutdown, but a queued job kept transmitting data.



Discussion: diagnosis

Give two different causal explanations consistent with this observation. What additional evidence would distinguish them?

Compare explanations before choosing one.

Discussion: constraint and test

Choose one explanation. Propose a shutdown requirement and a test that could reveal a violation. What would passing that test still leave unestablished?

Make the requirement and the observed endpoint agree.

Discussion: scaling oversight

The team plans to scale from a few agents to hundreds. It proposes using an LLM to generate hazard analyses or monitor incident logs for incomplete shutdowns. Choose one role. Describe how this could produce false assurance as the deployment scales. What evidence would you require before relying on it?

Choose one role: hazard analysis or incident monitoring.

Specifying, analyzing, and assuring differ

Activity	We defined a boundary and analyzed a scenario.
Observation	A hypothetical shutdown drill succeeded.
Constraint	Relevant work must cease within a justified deadline.
Conclusion	Requires evidence and an argument for the scoped claim.

Hazard analysis helps derive constraints. Assurance requires evidence that supports the intended claim.

Next: Module 2, model behavior. Scoped claims and proportionate evidence remain necessary.